Diagnosing Quadrant and Parameter-Orientation Errors

Focus: Quadrant-correct polar angles, negative-r conversions, parametric orientation, and complete parameter solutions.

Directions: Several problems show a student's work that is *wrong*. For those, (i) name the error in one sentence, then (ii) give the corrected value. Angles are in degrees unless a problem uses t in radians; give angles in $[0°, 360°)$ and round decimals to two places. A calculator's arctan always returns an angle in $(-90°, 90°)$ — that is where most of these errors begin.

1. Find and fix. Devin converts the point $P(-3, 4)$ to polar form. He writes $r = \sqrt{(-3)^2 + 4^2} = 5$, then $\theta = \arctan\left(\dfrac{4}{-3}\right) = -53.13°$, and reports $(5, -53.13°)$.

State what is wrong with his angle, then give the correct θ in $[0°, 360°)$.

Answer: ________________ degrees

2. Convert $Q(2, -2)$ to polar form. A classmate reasons: "the calculator gives $-45°$, and a negative angle is not allowed, so I add $180°$ and get $135°$."

Sketch the point, explain why $135°$ cannot be right, and give the correct θ in $[0°, 360°)$.

Answer: ________________ degrees

3. Find and fix. Sasha converts the polar point $(-4, 60°)$ to rectangular coordinates and reports $(2, 3.46)$.

Explain what a *negative* r does to the location of the point, then compute the correct y-coordinate.

Answer: ________________

4. A student claims that $x = 3\sin t$, $y = 3\cos t$ is "the same as" $x = 3\cos t$, $y = 3\sin t$, and evaluates the first at $t = \frac{\pi}{3}$ as $x = 1.5$.

Find the correct x-coordinate at $t = \frac{\pi}{3}$ for $x = 3\sin t$, $y = 3\cos t$, and say in one sentence how the two parametrizations differ in starting point and direction.

Answer: ________________

5. Find and fix. The ellipse $x = 4\cos t$, $y = 2\sin t$ is traced for $0^\circ \le t < 360^\circ$. Asked for every parameter value where the curve reaches $y = 1$, a student answers "$t = 30^\circ$" and stops.

Explain why one solution cannot be the whole answer for a curve that is traced all the way around, then list **all** such t in $[0^\circ, 360^\circ)$.

Answer: ________________ degrees

6. Find and fix. Converting $R(-5, -12)$ to polar form, a student computes $r = 13$ and $\theta = \arctan\left(\dfrac{-12}{-5}\right) = 67.38^\circ$, then reports $(13, 67.38^\circ)$.

Which quadrant does R actually lie in? Name the error, then give the correct θ in $[0^\circ, 360^\circ)$.

Answer: ________________ degrees

7. Rewrite the polar point $(-6, 200°)$ with a **positive** r and an angle in $[0°, 360°)$. A student writes $(6, 380°)$.

Explain why adding $180°$ was the wrong move here, then give the correct angle.

Answer: ________________ degrees

8. Synthesis (explain and sketch). Consider

$$C_1 : \; x = 2 + 3\cos t, \quad y = -1 + 3\sin t$$

$$C_2 : \; x = 2 + 3\cos(-t), \quad y = -1 + 3\sin(-t),$$

both for $0 \le t < 2\pi$.

Show that C_1 and C_2 are the same circle but are traced in opposite directions. Sketch the circle, mark the starting point, and put an arrow on each curve. Then explain how a small table of t values proves the orientation — and why the Cartesian equation alone never can.